Math 15 — Unit 5: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Vocabulary

1. Fill in each blank.

(a) The horizontal number line on a Cartesian plane is called the _____.

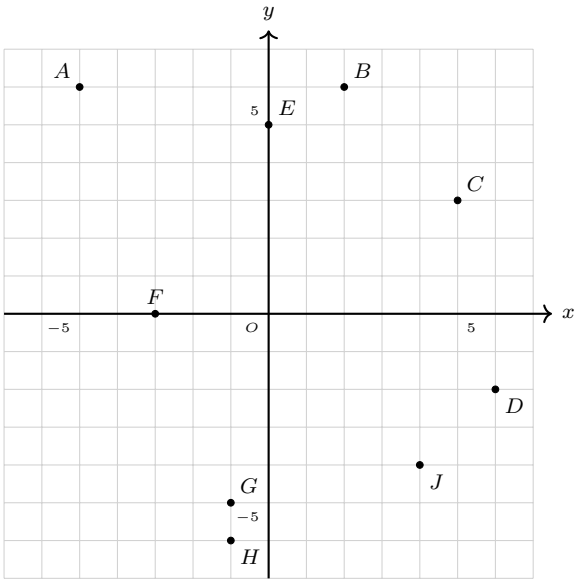
(b) The vertical number line on a Cartesian plane is called the _____.

(c) The point where the two axes cross, with coordinates $(0, 0)$, is called the _____.

(d) The two numbers used together to locate a point are called an _____ _____.

(e) The Cartesian plane is divided into four regions called _____.

Part B — Reading a Grid (Questions 2–7)



2. Name the point for each ordered pair.

i) $(-5, 6)$ ii) $(4, -4)$ iii) $(0, 5)$

3. List the coordinates of each point.

i) C ii) F iii) H iv) B

4. Which points lie in

i) Quadrant I ii) Quadrant II iii) Quadrant III iv) Quadrant IV

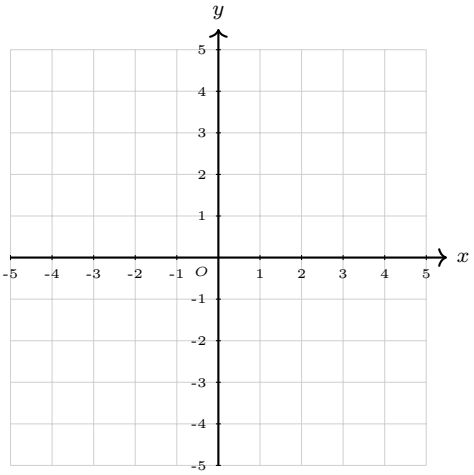
5. Which two points lie on an axis? For each one, name the axis it lies on.

6. Two of the points share the same y-coordinate. Which two points are they, and what does that tell you about where they are on the grid?

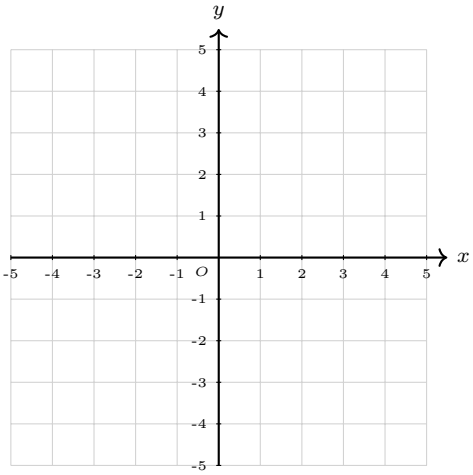
7. Two of the points share the same x-coordinate. Which two points are they, and what does that tell you about where they are on the grid?

Part C — Plotting and Joining Points

8. Plot the following points, in order, on the grid, and join each point to the next with a straight line segment: $(0,0) \to (5,0) \to (5,4) \to (0,4) \to (0,0)$. What geometric shape do the points outline? _____



9. Plot the following points on the grid, and join $(-3,-2)$ to $(3,-2)$, then $(3,-2)$ to $(0,3)$, then $(0,3)$ back to $(-3,-2)$. What geometric shape do the points outline? _____



Part D — Quadrants and Patterns

10. Without plotting, state which quadrant each point lies in. If a point lies on an axis, say which one.

(a) $(6,2)$ (b) $(-3,7)$ (c) $(-5,-1)$

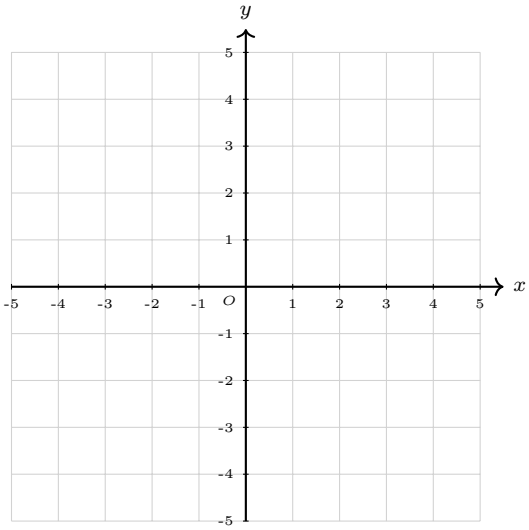
(d) $(2,-8)$ (e) $(0,-9)$ (f) $(7,0)$

11. Give one example of an ordered pair (using non-zero integers) that lies in each of the following quadrants.

(a) Quadrant I (b) Quadrant II (c) Quadrant III (d) Quadrant IV

12. Consider the points $(-3,-1)$, $(-1,1)$, $(1,3)$, and $(3,5)$.

(a) Plot the four points on the grid. (b) Do the points appear to lie on a straight line?



(c) Which quadrants do these four points lie in?

13. Without plotting it, explain how you know that the point $(-9,4)$ lies in Quadrant II.

Part E — Multiple Choice

14. Which of the following points lies in Quadrant III?
A. $(4, -2)$ B. $(-6, -3)$ C. $(5, 7)$ D. $(-2, 9)$

15. The point $(0, -6)$ lies
A. in Quadrant I. B. in Quadrant IV. C. on the x-axis. D. on the y-axis.

Answer Key. 1. (a) x-axis (b) y-axis (c) origin (d) ordered pair (e) quadrants. **2.** i) A ii) J iii) E. **3.** i) $C(5, 3)$ ii) $F(-3, 0)$ iii) $H(-1, -6)$ iv) $B(2, 6)$. **4.** i) B, C ii) A iii) G, H iv) D, J. **5.** $E(0, 5)$ on the y-axis; $F(-3, 0)$ on the x-axis. **6.** $A(-5, 6)$ and $B(2, 6)$ share $y = 6$ — same horizontal line. **7.** $G(-1, -5)$ and $H(-1, -6)$ share $x = -1$ — same vertical line. **8.** A rectangle (5 units by 4 units). **9.** A triangle. **10.** (a) I (b) II (c) III (d) IV (e) on the y-axis (f) on the x-axis. **11.** Answers vary. Sample: (a) $(1, 1)$ (b) $(-1, 1)$ (c) $(-1, -1)$ (d) $(1, -1)$. **12.** (b) Yes, the four points appear to lie on a straight line. (c) $(-3, -1)$ in III; $(-1, 1)$ in II; $(1, 3)$ and $(3, 5)$ in I. **13.** $x = -9$ is negative, so the point is left of the y-axis; $y = 4$ is positive, so it is above the x-axis — left and above is Quadrant II. **14.** B. **15.** D.

Lesson 2 Assignment *Show your work.*

Part A — Vocabulary and Review

1. Complete the following statements.

(a) The variable you choose or control is called the _____ variable.

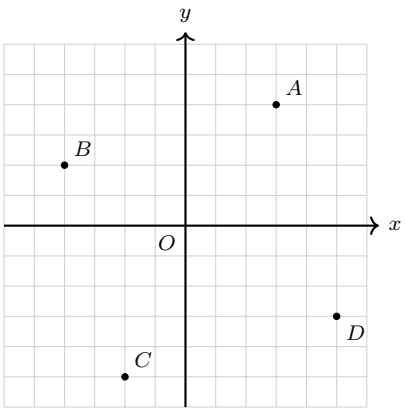
(b) The variable that results from, or depends on, the other variable is called the _____ variable.

(c) A table that organizes a pattern's input and output values is called a _____ _____.

(d) When graphing, the independent variable is plotted on the _____ axis.

(e) When graphing, the dependent variable is plotted on the _____ axis.

2. The grid at left shows four points. State the coordinates of points A, B, C, and D.



Part B — Patterns and Tables of Values

3. A pattern of tiles is built using bricks. Figure 1 uses 5 bricks, and each new figure uses 3 more bricks than the figure before it. (a) Complete the table relating the number of bricks, T, to the figure number, n.

Figure Number, n	1	2	3	4	5
Number of Bricks, T					

(b) How many bricks are added each time you move to the next figure?

(c) Continue the pattern to find the number of bricks in Figure 9.

4. A pattern of chairs is set up in rows for an assembly. Figure 1 has 8 chairs, and each new figure has 6 more chairs than the figure before it. (a) Complete the table relating the number of chairs, T, to the figure number, n.

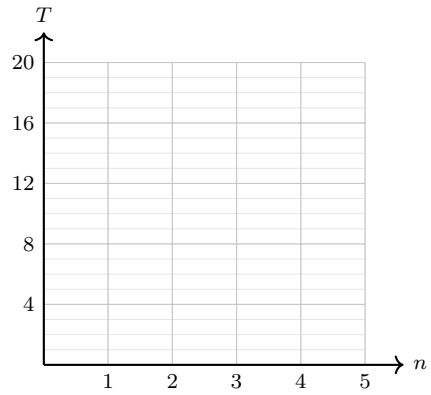
Figure Number, n	1	2	3	4	5
Number of Chairs, T					

(b) Describe in words how the number of chairs grows from one figure to the next.

(c) Continue the pattern to find the number of chairs in Figure 12.

5. A pattern of coins is stacked in piles. Figure 1 has 3 coins, and each new figure has 4 more coins than the figure before it. (a) Complete the table relating the number of coins, T, to the figure number, n, for Figures 1 through 5.

Figure Number, n	1	2	3	4	5
Number of Coins, T					



(b) Represent the data from the table on the grid at left.

(c) Do the plotted points appear to line up?

Part C — Independent and Dependent Variables

6. For each situation, identify the independent variable and the dependent variable.

(a) The number of pages read in a book, based on the number of minutes spent reading.

(b) The total amount saved in a piggy bank, based on the number of weeks spent saving.

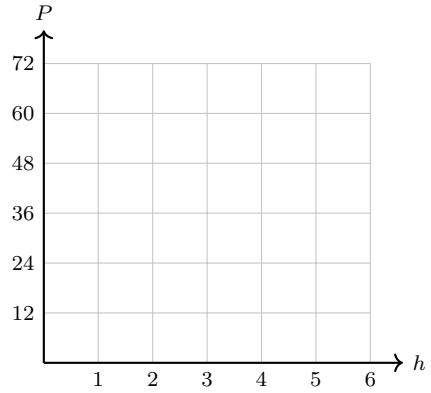
(c) The temperature of a glass of ice water, based on the number of minutes it sits on the counter.

(d) The total number of stickers given out, based on the number of students in the class.

7. The table below relates the number of hours worked, h, to the total pay, P dollars, for a part-time job.

Hours Worked, h	1	2	3	4	5
Pay (\$), P	12	24	36	48	60

(a) Identify the independent variable and the dependent variable.



(b) Represent the data from the table on the grid at left.

(c) Continue the pattern in the table to find the pay for 6 hours of work.

Part D — Short Answer

8. In your own words, explain the difference between an independent variable and a dependent variable. Then give your own real-life example of a pair of variables (not used elsewhere in this lesson), and state which one is independent and which one is dependent.

9. Explain how you can use a table of values to check whether a pattern is growing by a constant amount.

Part E — Multiple Choice and Numerical Response

10. *(Multiple Choice)* The total cost of filling a car with gas depends on the number of litres of gas pumped. Which variable is the *independent* variable?
A. the total cost B. the number of litres pumped
C. the type of car D. the price per litre

11. *(Multiple Choice)* A pattern begins at 4 and grows by 5 each figure. What is the value at Figure 6?
A. 24 B. 26 C. 29 D. 34

12. *(Multiple Choice)* When a table of values is graphed on the Cartesian plane, the dependent variable is always plotted on which axis?
A. the x-axis B. the y-axis
C. either axis D. neither axis

13. *(Numerical Response)* A pattern of tile figures starts at 6 tiles in Figure 1 and grows by 4 tiles each figure. The number of tiles in Figure 7 is _____.
Record your answer from left to right:

Answer Key. 1. (a) independent (b) dependent (c) table of values (d) horizontal (x) (e) vertical (y). **2.** $A(3,4)$, $B(-4,2)$, $C(-2,-5)$, $D(5,-3)$. **3.** (a) T: $5, 8, 11, 14, 17$ (b) 3 bricks (c) 29 bricks. **4.** (a) T: $8, 14, 20, 26, 32$ (b) starts at 8 and increases by 6 each figure (c) 74 chairs. **5.** (a) T: $3, 7, 11, 15, 19$ (b) plot $(1,3), (2,7), (3,11), (4,15), (5,19)$ (c) yes, they line up. **6.** (a) Ind: minutes reading; Dep: pages read (b) Ind: weeks saving; Dep: amount saved (c) Ind: minutes on counter; Dep: temperature (d) Ind: number of students; Dep: stickers given out. **7.** (a) Ind: hours worked h; Dep: pay P (b) plot $(1,12), (2,24), (3,36), (4,48), (5,60)$ (c) \$72. **8.** Answers vary: the independent variable is chosen or controlled; the dependent variable results from it. **9.** If the dependent values change by the same amount each time the independent variable increases by 1, the growth is constant. **10.** B. **11.** C. **12.** B. **13.** 30.

Lesson 3 Assignment *Show your work.*

Part A — Ordered Pairs and Tables of Values

1. The total price, P (dollars), of gasoline pumped is related to the number of litres, L, pumped. State (i) the independent variable and (ii) the dependent variable.

2. Movie tickets cost \$6 each. Let n be the number of tickets bought and C the total cost, in dollars. Write this relation as a set of ordered pairs (n, C) for $n = 1, 2, 3, 4$.

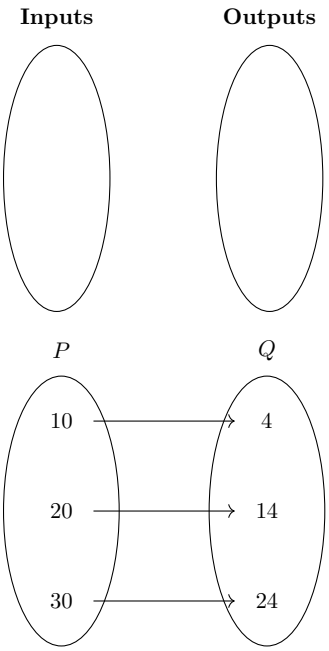
3. In a 25 m pool, the table shows the number of laps, l, a swimmer completes and the total distance, d (metres), swum. Write this relation as a set of ordered pairs (l, d).

l	2	4	6	8
d	50	100	150	200

4. A relation is given by the ordered pairs (x, y): $(1, 9),\ (2, 11),\ (3, 13),\ (4, 15)$. Complete a table of values for this relation.

x				
y				

Part B — Mapping Diagrams



5. A relation is given by the set of ordered pairs (m, n): $(3, 7),\ (5, 11),\ (7, 15)$. Represent this relation with the mapping diagram at left. Write each input in the left oval and its output in the right oval, then draw the arrows.

6. The mapping diagram at left shows a relation between two sets of numbers, P and Q.

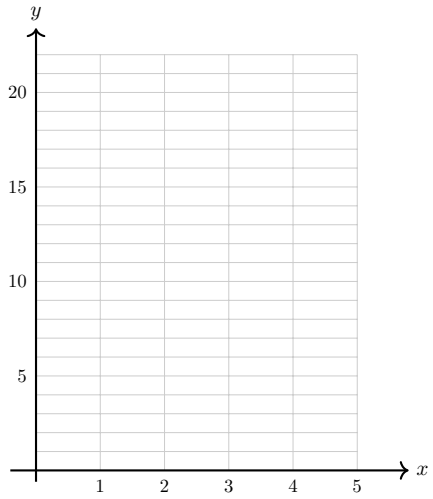
(a) Write this relation as a set of ordered pairs (P, Q).

(b) Complete a table of values for this relation.

P			
Q			

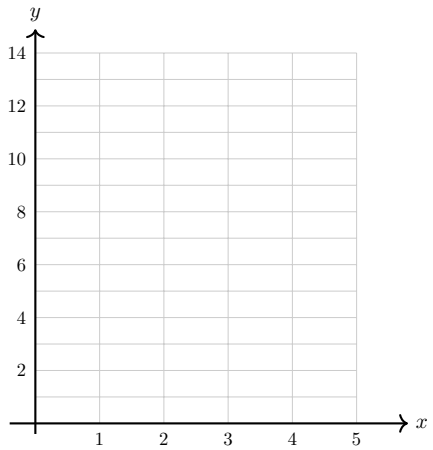
Part C — Graphing Relations

7. Plot these ordered pairs (x, y) on the grid below:
$(1, 5),\ (2, 10),\ (3, 15),\ (4, 20)$.

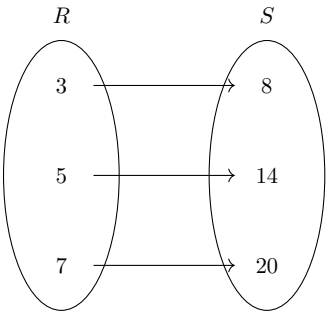


8. The table below shows a relation. Plot the corresponding ordered pairs on the grid below.

x	0	1	2	3	4
y	1	4	7	10	13



9. The mapping diagram below shows a relation between two sets of numbers, R and S.

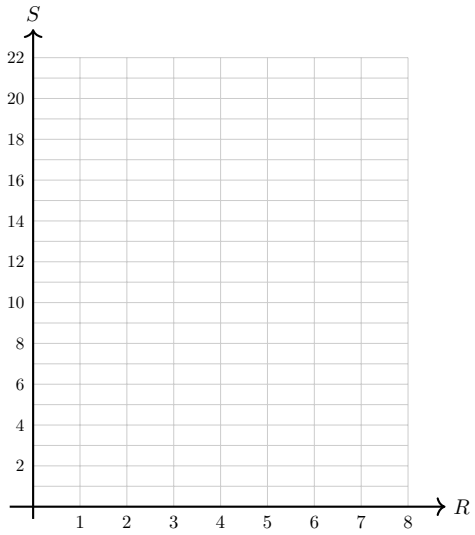


(a) Write this relation as a set of ordered pairs (R, S).

(b) Complete a table of values for this relation.

R			
S			

9. (c) Plot the ordered pairs from part (a) on the grid below to represent this relation as a graph.



Part D — Applications and Practice

10. A trivia night charges \$8 per person as a team's entry fee. Let p be the number of people on a team and F the team's total entry fee, in dollars.

(a) Determine the total entry fee for $p = 2, 3, 4$, and 5 people.

(b) Write this relation as a set of ordered pairs (p, F).

11. *(Multiple Choice)* In a mapping diagram, arrows are drawn from the
A. outputs to the inputs B. inputs to the outputs
C. x-axis to the y-axis D. dependent variable to the independent variable

12. *(Multiple Choice)* For the relation $x : 1, 2, 3, 4$ and $y : 6, 9, 12, 15$, which ordered pair does **not** belong?
A. $(1, 6)$ B. $(2, 9)$ C. $(3, 10)$ D. $(4, 15)$

13. *(Numerical Response)* The table below shows a relation.

x	1	2	3	4	5
y	4	9	14	19	___

(a) Determine the missing output value. _____

(b) Determine the sum of all five outputs in the completed table. _____

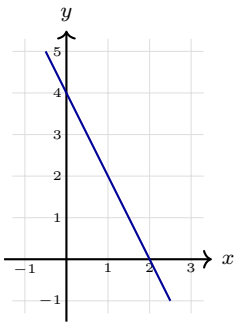
Answer Key. 1. (i) independent L (ii) dependent P. **2.** $(1,6),(2,12),(3,18),(4,24)$. **3.** $(2,50),(4,100),(6,150),(8,200)$. **4.** $x : 1,2,3,4$; $y : 9,11,13,15$. **5.** inputs $3,5,7$ (left oval), outputs $7,11,15$ (right oval); arrows $3 \to 7$, $5 \to 11$, $7 \to 15$. **6.** (a) $(10,4),(20,14),(30,24)$ (b) $P : 10,20,30$; $Q : 4,14,24$. **7.** points at $(1,5),(2,10),(3,15),(4,20)$. **8.** points at $(0,1),(1,4),(2,7),(3,10),(4,13)$. **9.** (a) $(3,8),(5,14),(7,20)$ (b) $R : 3,5,7$; $S : 8,14,20$ (c) those three points plotted. **10.** (a) $16,24,32,40$ (b) $(2,16),(3,24),(4,32),(5,40)$. **11.** B. **12.** C. **13.** (a) 24 (b) 70.

Lesson 4 Assignment *Show your work.*

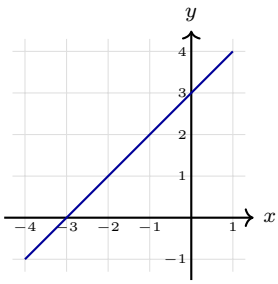
Part A — Reading Intercepts from Graphs

1. Fill in each blank.
(a) At an x-intercept, the value of y is _____
(b) At a y-intercept, the value of x is _____
(c) An x-intercept has the form (_____, 0)
(d) A y-intercept has the form (0, _____)

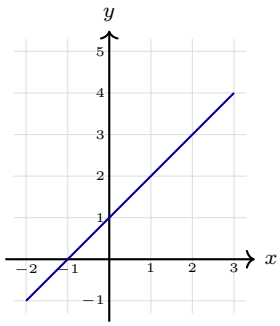
2. State the x-intercept of each graph.
2(a) _____



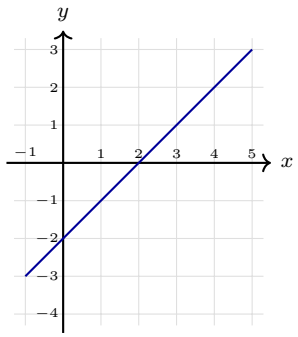
2(b) _____



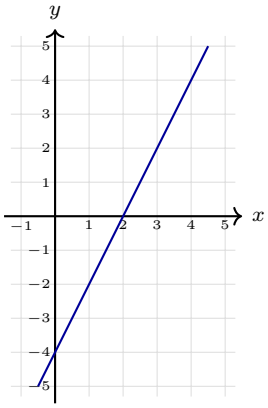
3. State the y-intercept of each graph.
3(a) _____



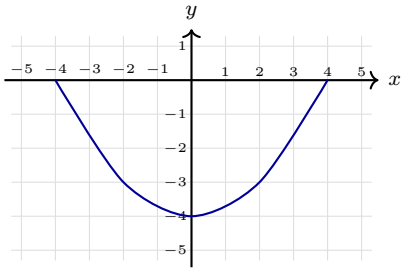
3(b) _____



4. The graph below shows a line. State the x-intercept and the y-intercept.



5. The curve below crosses the x-axis at two points and the y-axis at one point. State all three intercepts.



Part B — Intercepts from Tables, and Key Facts

6. The table shows several points on a relation. State the x-intercept and the y-intercept.

x	-3	-2	-1	0	1
y	-2	-1	0	1	2

7. The table shows several points on a relation, listed out of order. State the x-intercept and the y-intercept.

x	2	-1	4	0	3
y	0	-9	6	-6	3

8. State whether each sentence is **True** or **False**.
(a) At an x-intercept, the y-coordinate is always 0.
(b) At a y-intercept, the y-coordinate is always 0.
(c) A graph can have more than one x-intercept.
(d) $(0, 0)$ could be both the x- and the y-intercept of a graph.
(e) If a graph passes through $(5, 0)$, then 5 is the y-intercept.

Part C — Applications and Extra Practice

9. A candle is 20 cm tall and burns down at a constant rate of 2 cm per hour. The table shows the height, h cm, of the candle after t hours.

t (h)	0	2	4	6	8	10
h (cm)	20	16	12	8	4	0

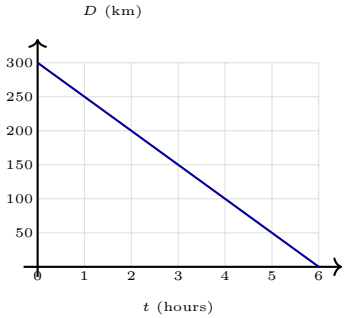
(a) What is the y-intercept (the height-intercept)? What does it represent?

(b) What is the x-intercept (the time-intercept)? What does it represent?

(c) Using the burn rate, determine the height of the candle after 5 hours.

(d) Is this relation linear or nonlinear? Explain how you know.

10. Priya is on a road trip. The graph shows the distance remaining, D km, until she reaches her destination, after t hours of driving.



(a) What is the y-intercept of the graph? What does it represent?

(b) What is the x-intercept of the graph? What does it represent?

(c) Use the graph to estimate the distance remaining after 2 hours.

(d) Use the graph to estimate after how many hours only 100 km remain.

11. *(Multiple Choice)* Which ordered pair could be the x-intercept of a graph?
A. $(0, 7)$ B. $(-3, 0)$ C. $(2, 5)$ D. $(0, -1)$

12. *(Multiple Choice)* Which ordered pair could be the y-intercept of a graph?
A. $(6, 0)$ B. $(0, -9)$ C. $(-2, -2)$ D. $(4, 0)$

13. A student says, "A straight line can cross the x-axis at two different points, just like the curve in Question 5." Is the student correct? Explain your answer. (*Hint:* try to draw one.)

Answer Key. 1. (a) 0 (b) 0 (c) any number (d) any number. **2.** (a) $(2, 0)$ (b) $(-3, 0)$. **3.** (a) $(0, 1)$ (b) $(0, -2)$. **4.** x-int $(2, 0)$, y-int $(0, -4)$. **5.** x-ints $(-4, 0)$ and $(4, 0)$, y-int $(0, -4)$. **6.** x-int $(-1, 0)$, y-int $(0, 1)$. **7.** x-int $(2, 0)$, y-int $(0, -6)$. **8.** (a) True (b) False (c) True (d) True (e) False. **9.** (a) $(0, 20)$; the candle's starting height (b) $(10, 0)$; the time the candle is completely burned out (c) 10 cm (d) linear; the height drops by the same 2 cm each hour. **10.** (a) $(0, 300)$; the total distance of the trip before any driving (b) $(6, 0)$; after 6 h, 0 km remain — Priya has arrived (c) about 200 km (d) about 4 hours. **11.** B. **12.** B. **13.** No; a non-horizontal straight line never bends back, so it can cross the x-axis at most once (unless the line lies along the x-axis itself). Only a curve that turns around can cross it more than once.

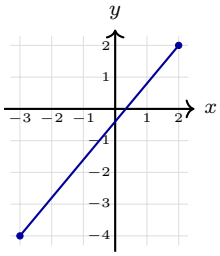
Lesson 5 Assignment *Show your work.*

Part A — Reading Domain and Range From a Graph

1. Fill in each blank.
(a) The domain of a relation is the set of all possible _____-values.
(b) The range of a relation is the set of all possible _____-values.
(c) A discrete graph is made up of separate _____.
(d) A continuous graph is an unbroken _____.

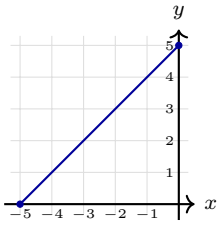
2. State the domain of each graph below.

2(a)



D: _____

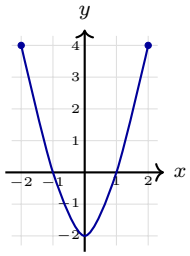
2(b)



D: _____

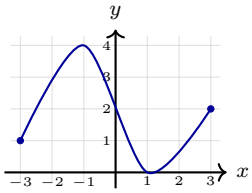
3. State the range of each graph below.

3(a)



R: _____

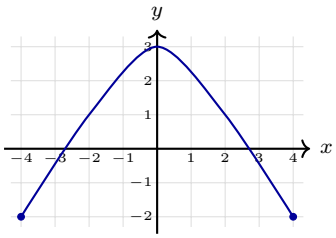
3(b)



R: _____

4. The graph shows a curve; state its domain and range.

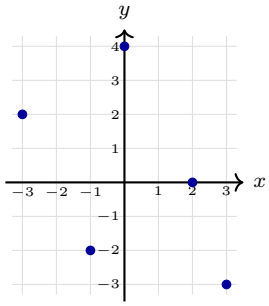
4.



D: _____
R: _____

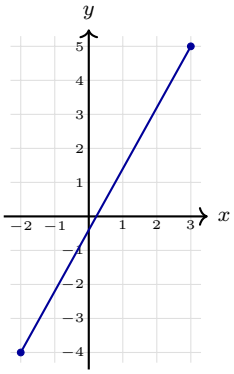
5. For each graph, state whether it is discrete or continuous, then state its domain and range.

5(a)



Type: _____
D: _____
R: _____

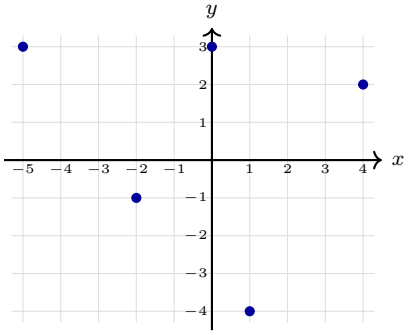
5(b)



Type: _____
D: _____
R: _____

6. The graph shows a set of discrete points; state the domain and range.

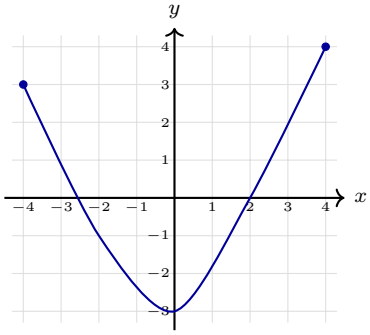
6.



D: _____ R: _____

7. The graph shows a continuous curve; state the domain and range. (Hint: look at the whole curve, not just the two ends.)

7.



D: _____ R: _____

Part B — Everyday Graphs and Extra Practice

8. State whether each sentence is **True** or **False**.

(a) The domain of a relation is the set of all y-values. _____

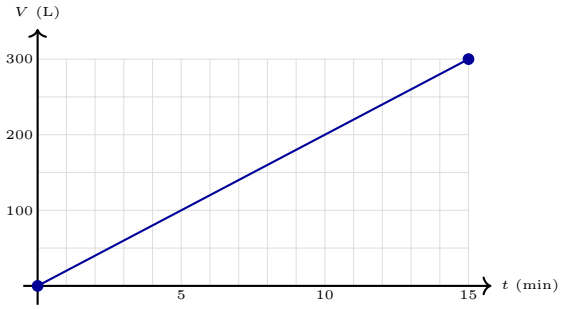
(b) A discrete graph is described using a list of values, not "from __ to __." _____

(c) The range of a graph can be found by looking at the lowest and highest points on the graph. _____

(d) Every continuous graph goes on forever in both directions. _____

(e) A graph's domain could include only positive numbers. _____

9. A garden hose fills a small pool. The graph shows the volume of water, V litres, in the pool after t minutes. The pool starts empty and fills steadily, reaching 300 L after 15 minutes.

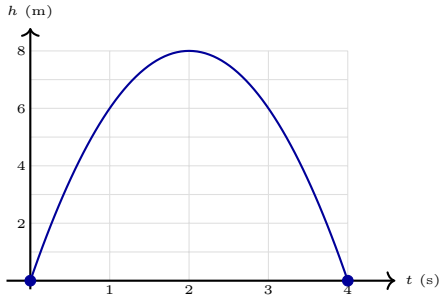


(a) Domain (in minutes): _____

(b) Range (in litres): _____

(c) Explain why the domain could not include negative values of t.

10. A soccer ball is kicked into the air. The graph shows the ball's height, h metres, after t seconds. It peaks at 8 m when $t = 2$ s and lands at $t = 4$ s.



(a) Domain (in seconds): _____

(b) Range (in metres): _____

(c) Explain why the domain could not include negative values of t.

11. *(Multiple Choice)* A graph is made up of four separate dots, at $x = -5,\ -1,\ 2$, and 6. Which best describes how to state its domain?

A. x can be any value from -5 to 6. B. x can be $-5,\ -1,\ 2$, or 6.

C. The domain and the range are equal. D. The domain is all real numbers.

12. *(Numerical Response)* A discrete relation includes the ordered pairs $(-6, 5)$, $(-3, 9)$, $(0, 5)$, $(2, 14)$, $(5, 9)$. If a is the number of elements in the domain and b is the number of elements in the range, the value of $a+b$ is _____.

13. An oven starts at room temperature, 20°C, and preheats to 200°C over the first 8 minutes after being turned on. After that, the oven stays at a constant 200°C. Consider only the graph of the first 8 minutes (the preheating part).

(a) State the domain of this part of the graph (in minutes).

(b) State the range of this part of the graph (in degrees Celsius).

(c) Is this part of the graph discrete or continuous? Explain.

Answer Key. **1.** (a) x (b) y (c) dots/points (d) line or curve. **2.** (a) x from -3 to 2 (b) x from -5 to 0. **3.** (a) y from -2 to 4 (b) y from 0 to 4. **4.** D: -4 to 4; R: -2 to 3. **5.** (a) discrete; D: $-3, -1, 0, 2, 3$; R: $-3, -2, 0, 2, 4$ (b) continuous; D: -2 to 3; R: -4 to 5. **6.** D: $-5, -2, 0, 1, 4$; R: $-4, -1, 2, 3$. **7.** D: -4 to 4; R: -3 to 4. **8.** (a) False (b) True (c) True (d) False (e) True. **9.** (a) 0 to 15 min (b) 0 to 300 L (c) negative time would mean before the hose was turned on, which is not part of this filling situation. **10.** (a) 0 to 4 s (b) 0 to 8 m (c) negative time would mean before the ball was kicked, which does not apply to this flight. **11.** B. **12.** 8. **13.** (a) 0 to 8 min (b) 20°C to 200°C (c) continuous; the temperature changes smoothly with no gaps or jumps as the oven heats up.

Lesson 6 Assignment *Show your work.*

Part A — Ordered Pairs and Tables

1. Fill in each blank.
(a) The domain of a relation is the set of all _____ values.
(b) The range of a relation is the set of all _____ values.
(c) In (x, y), the domain value is the _____-coordinate.
(d) In (x, y), the range value is the _____-coordinate.

2. List the domain and range of each set of ordered pairs.
(a) $(2, 6),\ (3, 6),\ (5, 9),\ (7, 1)$
(b) $(-4, 3),\ (-4, 7),\ (-4, 0)$

3. State the domain and range of each relation given by a table of values.
(a)

x	-2	-1	0	1	2
y	9	7	5	3	1

(b)

x	0	2	4	6
y	8	3	8	3

4. State the domain and range of the relation below. Notice the x-values are not listed in order.

x	3	-1	4	0	2
y	9	1	16	0	4

5. Suppose the x-coordinates of the ordered pairs in a relation are 4, 4, 4, 9, 9, 2. Write the domain of this relation using proper set notation.

Part B — Set Notation and Inequalities

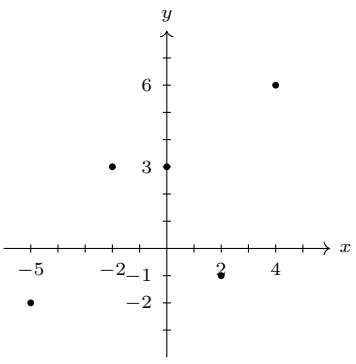
6. Write the domain and range, using set notation, of the ordered pairs $(5, 2),\ (5, 9),\ (5, -3),\ (5, 0)$.

7. Write the domain and range, using set notation, of the relation given by the table below.

x	10	20	30	40	50
y	100	100	100	100	100

8. Each description below is part of a graph that continues forever in one direction, like the graphs from Lesson 5. Write the indicated set using an inequality.
(a) A ray starts at the point $(1, 4)$ and rises forever to the upper right, with an arrow at the end. Write the **range**.
(b) A ray starts at the point $(6, 2)$ and extends forever to the left, with an arrow at the end. Write the **domain**.

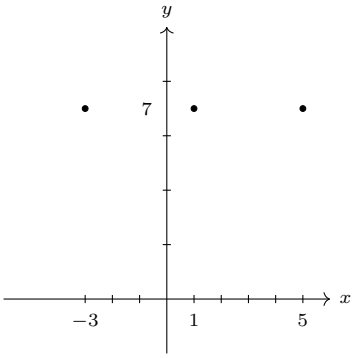
Part C — Combining Representations and Applications



9. State the domain and range, using set notation, of the relation shown in the graph at the left.

10. A relation is given below in three different forms. Determine the domain and range for each, writing your answers in set notation.

(a) Ordered pairs: $(1, 10),\ (2, 20),\ (3, 10),\ (4, 40)$



(b) Table:

x	-6	-3	0	3
y	-1	-1	2	5

(c) Graph: the one shown at the left.

11. A snack bar sells popcorn combos for \$4 each during a Saturday afternoon shift. Only 60 combos are available that shift. The table shows the number of combos sold, n, and the total revenue, R dollars, collected.

n	10	20	30	40	50	60
R	40	80	120	160	200	240

(a) State the domain and range of the relation shown in the table, using set notation.

(b) Explain why n cannot be greater than 60 in this situation.

(c) Could the snack bar collect \$260 from popcorn combos during this shift? Explain.

Part D — Concepts and Review

12. State whether each sentence is **True** or **False**.

(a) The domain of a relation can never contain a repeated value once written in set notation.

(b) If two ordered pairs share the same y-value, that value is listed twice in the range.

(c) The order in which values are listed inside curly braces changes the set.

(d) The inequality $y \geq 0$ could describe the range of a relation whose graph continues upward forever, starting from a lowest point of $y = 0$.

(e) A relation with exactly one ordered pair has a domain with one element and a range with one element.

13. *(Multiple Choice)* According to the set notation convention used in this lesson (each value listed once), which of the following correctly shows the **range** of the ordered pairs $(0, 3),\ (1, 3),\ (2, 8),\ (3, 3)$? A. $\{0, 1, 2, 3\}$ B. $\{3, 3, 3, 8\}$ C. $\{3, 8\}$ D. $\{8\}$

14. *(Numerical Response)* A relation includes the ordered pairs $(-6, 14),\ (-2, 9),\ (0, 9),\ (3, 14),\ (5, 20)$. When the range is written in set notation, the sum of all the numbers in the range is _____.

15. In your own words, explain the difference between writing a domain as a list in curly braces (such as $\{2, 4, 6\}$) and writing a domain using an inequality (such as $x \geq 0$). Give one example of a relation where each type of notation would be the better choice.

Answer Key. 1. (a) input (b) output (c) x (d) y. **2.** (a) $D: 2, 3, 5, 7$; $R: 1, 6, 9$ (b) $D: -4$; $R: 0, 3, 7$. **3.** (a) $D: -2, -1, 0, 1, 2$; $R: 1, 3, 5, 7, 9$ (b) $D: 0, 2, 4, 6$; $R: 3, 8$. **4.** $D: -1, 0, 2, 3, 4$; $R: 0, 1, 4, 9, 16$. **5.** $D = \{2, 4, 9\}$. **6.** $D = \{5\}$, $R = \{-3, 0, 2, 9\}$. **7.** $D = \{10, 20, 30, 40, 50\}$, $R = \{100\}$. **8.** (a) $y \geq 4$ (b) $x \leq 6$. **9.** $D = \{-5, -2, 0, 2, 4\}$, $R = \{-2, -1, 3, 6\}$. **10.** (a) $D = \{1, 2, 3, 4\}$, $R = \{10, 20, 40\}$ (b) $D = \{-6, -3, 0, 3\}$, $R = \{-1, 2, 5\}$ (c) $D = \{-3, 1, 5\}$, $R = \{7\}$. **11.** (a) $D = \{10, 20, 30, 40, 50, 60\}$, $R = \{40, 80, 120, 160, 200, 240\}$ (b) only 60 combos are available that shift, so it is impossible to sell more than 60 (c) no; \$260 would need 65 combos, and 65 combos can't be sold when only 60 exist (the most is \$240). **12.** (a) True (b) False (c) False (d) True (e) True. **13.** C. **14.** 43. **15.** A list in braces is used for a *finite, discrete* relation where every value can be written down (e.g. $D = \{2, 4, 6\}$ for $(2, 1), (4, 1), (6, 1)$); an inequality is used for a *continuous* relation that keeps going forever, where listing every value is impossible (e.g. $x \geq 0$ for a ray starting at the origin and extending infinitely right).

Lesson 7 Assignment *Show your work.*

Part A — Vocabulary and Choosing a Window

1. Fill in each blank.

(a) The set of all possible x-values (inputs) of a relation is called the _____.

(b) The set of all possible y-values (outputs) of a relation is called the _____.

(c) The setting that controls the smallest and largest x- and y-values shown on a screen is called the viewing _____.

(d) A point where a graph crosses the x-axis is called an _____.

(e) A point where a graph crosses the y-axis is called a _____.

2. Without graphing, determine the x- and y-intercepts of each relation, and decide whether the **standard window** ($x : [-10, 10]$, $y : [-10, 10]$) would show both intercepts or would need to be adjusted.

(a) $y = x + 3$

(b) $y = 2x - 5$

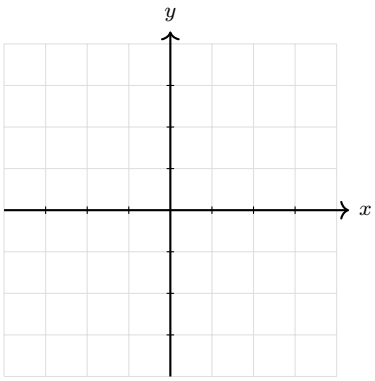
(c) $y = 6x - 42$

(d) $y = -0.5x + 3$

(e) $y = -8x - 72$

(f) $y = 3x + 27$

3. Consider the relation $y = 4x + 4$.



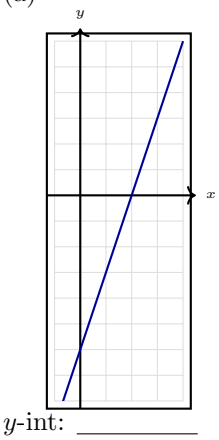
(a) Use a graphing calculator or Desmos to graph the relation. State a window that shows both intercepts clearly.

(b) Sketch the graph on the axes at the left.

Part B — Reading Intercepts, Domain, and Range Off a Screen

4. Using the graphs below (as they would appear on a calculator or Desmos screen), state the x-intercept and the y-intercept of each relation. The gridlines are 1 unit apart.

(a) x-int: _____



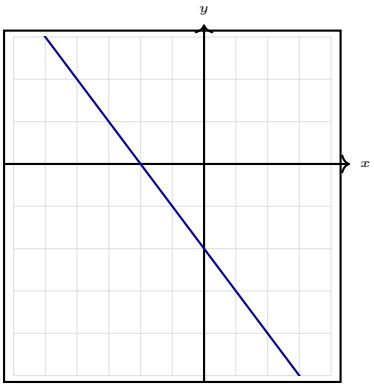
y-int: _____

(b) x-int: _____



y-int: _____

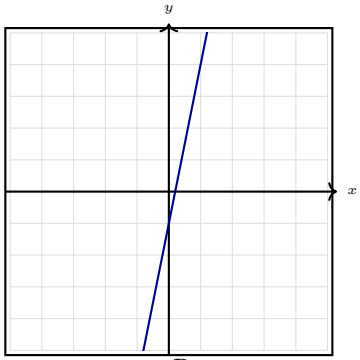
5. The screen shows $y = -2x - 4$ using the window $x : [-6, 4]$, $y : [-10, 6]$.



Use the graph, or the calculator's zero/table/value features, to state the x-intercept and the y-intercept.

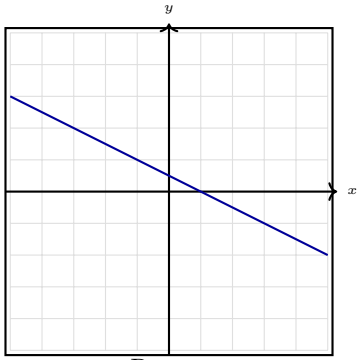
6. Each screen below shows a straight, slanted line in the standard window. For each one, state the domain and the range.

(a) $y = 5x - 2$



Domain: _____ Range: _____

(b) $y = -0.5x + 1$



Domain: _____ Range: _____

7. State whether each sentence is **True** or **False**.

(a) A graphing calculator's window controls which part of a graph is visible on the screen.

(b) If a relation's y-intercept lies outside the window's Ymin/Ymax range, the graph will not show it.

(c) The domain of every straight, non-horizontal line is a single number.

(d) The trace feature can be used to find the y-value at a given x-value.

(e) Adjusting the window changes the actual shape of the relation being graphed.

Part C — Applications and Extra Practice

8. A dog-walking business charges a \$6 booking fee plus \$4 per walk. The total cost, C dollars, for n walks booked in a month is $C = 6 + 4n$.

(a) Use technology to graph this relation. State a window that is appropriate for this situation.

(b) Determine the C-intercept using technology, and explain what it means in this situation.

(c) Determine the cost of booking 5 walks.

(d) If a customer paid \$50, how many walks did they book?

(e) State an appropriate domain and range for this relation, given the context.

9. A rental company charges a flat \$12 fee plus \$3 per hour to rent a canoe. The total cost, C dollars, to rent the canoe for h hours is $C = 12 + 3h$.

(a) Use technology to graph this relation. State a reasonable window.

(b) Determine the C-intercept using technology, and explain what it means in this situation.

(c) Algebraically determine the h-intercept. Does this value make sense in this situation? Explain.

(d) Determine the cost to rent the canoe for 4 hours.

(e) If the total cost was \$30, how many hours was the canoe rented?

(f) State the domain and range of this relation in context. (*Hint:* a canoe can be rented for a fraction of an hour.)

10. *(Multiple Choice)* A calculator's standard window is $x : [-10, 10]$, $y : [-10, 10]$. Which relation below would have its y-intercept cut off (not visible) in that window? A. $y = 2x + 5$ B. $y = -3x + 40$ C. $y = 0.5x - 8$ D. $y = 4x + 2$

11. *(Multiple Choice)* Which statement about the domain and range of a straight, non-horizontal, non-vertical line is true? A. The domain is always positive numbers only. B. The domain and range are both all real numbers. C. The range depends only on the y-intercept. D. A line has no domain or range.

12. Consider the relation $y = -4x + 20$.

(a) Determine the x-intercept and the y-intercept using technology (or algebra).

(b) State a window that would show both intercepts clearly.

(c) State the domain and range of this relation.

(d) Suppose this relation instead represented the battery charge remaining, y%, on a phone after x hours of use. Would the domain and range from part (c) still make sense? Explain, and state a more realistic domain and range.

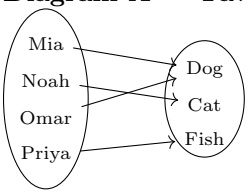
Answer Key. 1. (a) domain (b) range (c) window (d) x-intercept (e) y-intercept. **2.** (a) x-int -3, y-int 3; standard OK (b) 2.5, -5; OK (c) 7, -42; adjust (d) 6, 3; OK (e) -9, -72; adjust (f) -9, 27; adjust. **3.** (a) varies, e.g. $x : [-3, 3]$, $y : [-4, 8]$ (b) line through $(-1, 0)$ and $(0, 4)$. **4.** (a) x-int 2, y-int -6 (b) x-int 4, y-int 8. **5.** x-int -2, y-int -4. **6.** (a) and (b) domain: all real numbers; range: all real numbers. **7.** (a) T (b) T (c) F (d) T (e) F. **8.** (a) varies, e.g. $x : [0, 12]$, $y : [0, 55]$ (b) C-int $= 6$: the \$6 booking fee charged even for 0 walks (c) \$26 (d) 11 walks (e) domain $\{0, 1, 2, 3, \dots\}$; range $\{6, 10, 14, 18, \dots\}$ ($C = 6 + 4n$). **9.** (a) varies, e.g. $x : [0, 10]$, $y : [0, 45]$ (b) C-int $= 12$: the flat \$12 fee charged even for 0 hours (c) h-int $= -4$; not meaningful — rental time cannot be negative (d) \$24 (e) 6 hours (f) domain $h \geq 0$, $h \in R$; range $C \geq 12$, $C \in R$. **10.** B. **11.** B. **12.** (a) x-int 5, y-int 20 (b) varies, e.g. $x : [-1, 7]$, $y : [-5, 25]$ (c) domain and range both all real numbers (d) No — hours of use cannot be negative and battery % cannot fall below 0; realistic domain $0 \leq x \leq 5$, range $0 \leq y \leq 20$.

Lesson 8 Assignment *Show your work.*

Part A — Is It a Function?

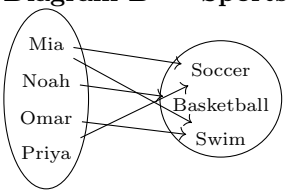
1. Diagram A shows each student's *one* favourite pet; Diagram B shows every sport each student plays. Is each relation a function? Explain.

Diagram A — Favourite Pet



(a) _____ Explain:

Diagram B — Sports Played



(b) _____ Explain:

2. Determine whether each relation is a function. Give a reason for each answer.
(a) $(2,5)$, $(3,5)$, $(4,5)$, $(6,5)$ (b) $(1,3)$, $(1,7)$, $(2,9)$, $(4,10)$ (c) $(0,0)$, $(1,1)$, $(2,4)$, $(3,9)$ (d) $(5,2)$, $(5,2)$, $(6,3)$, $(7,4)$

3. Determine whether each table represents a function.

(a)

x	1	2	3	4
y	10	20	30	40

(b)

x	3	5	3	8
y	1	2	4	6

4. State whether each relation described in words is a function, and explain.
(a) Each licensed driver in Alberta paired with their driver's licence number.

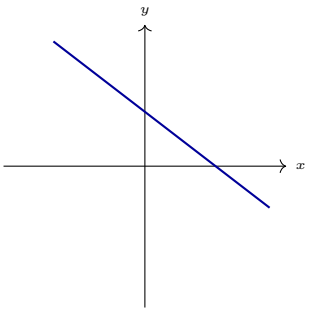
(b) Each ice cream flavour sold at a shop paired with the students who chose it as their favourite.

5. Fill in the blanks: a function is a relation in which every _____ is paired with exactly _____ output.

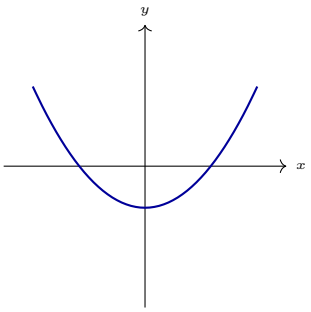
6. In your own words, state the vertical line test.

Part B — The Vertical Line Test

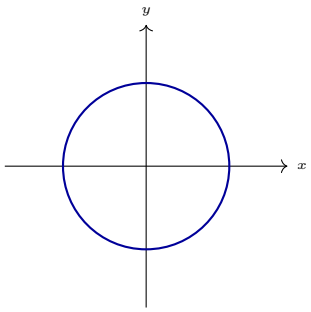
7. Use the vertical line test to classify each graph as a function or not a function.



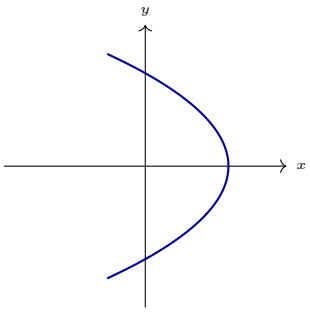
(a) _____



(b) _____



(c) _____

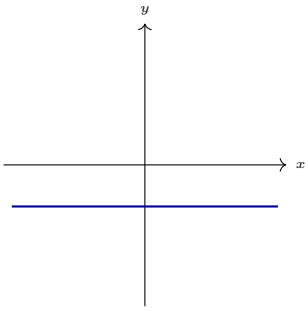


(d) _____

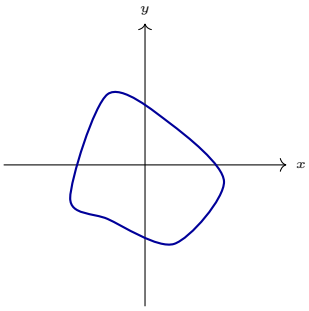
8. From the graph in Question 7(c) (the circle), write two ordered pairs that share the same x-value but have different y-values, to show why it is not a function.

9. From the graph in Question 7(d) (the sideways curve), write two ordered pairs that share the same x-value but have different y-values, to show why it is not a function.

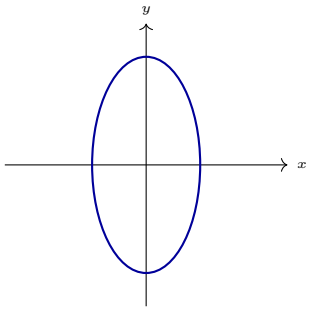
10. Use the vertical line test to classify each graph below (write Yes or No).



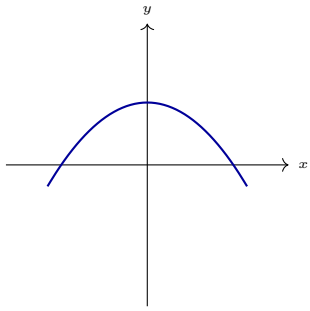
E: _____



F: _____



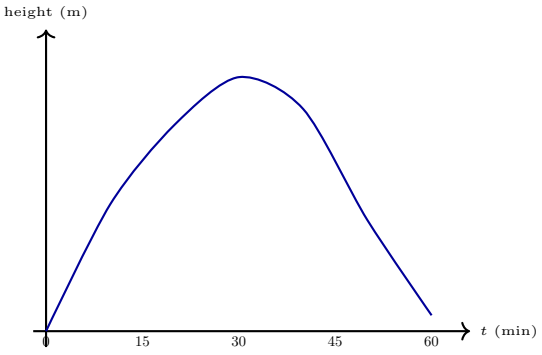
G: _____



H: _____

Part C — Applications and Review

11. The graph shows the height of a hot-air balloon, in metres, recorded every few minutes during a one-hour flight.



Is height-over-time a function? Explain your reasoning using the vertical line test.

12. Consider the relation that pairs each test score earned on a math test with the students in the class who earned that score. Is this relation a function? Explain.

13. *(Multiple Choice)* Which set of ordered pairs is **not** a function?
A. $(1,2)$, $(2,3)$, $(3,4)$, $(4,5)$ B. $(0,5)$, $(1,5)$, $(2,5)$, $(3,5)$
C. $(2,1)$, $(2,4)$, $(3,6)$, $(4,8)$ D. $(-1,0)$, $(0,1)$, $(1,2)$, $(2,3)$

14. State whether each statement is true (T) or false (F).
(a) A function can have two different outputs for the same input. _____
(b) Every vertical line drawn through the graph of a function crosses it at most once. _____
(c) A relation containing only one ordered pair is always a function. _____

x	2	4	6	8
y	9	9	9	9

15. The table at left shows a relation. Is it a function, even though every output is the same number? Explain why or why not.

Answer Key. 1. (a) Function — each student has exactly one arrow (b) Not a function — Mia has two arrows. **2.** (a) Function (b) Not a function ($x = 1$ twice) (c) Function (d) Function. **3.** (a) Function (b) Not a function ($x = 3$ gives two different outputs). **4.** (a) Function (b) Not a function. **5.** input; one. **6.** If any vertical line crosses the graph more than once, the relation is not a function. **7.** (a) Function (b) Function (c) Not a function (d) Not a function. **8.** $(0,2)$ and $(0,-2)$. **9.** $(1.6,1)$ and $(1.6,-1)$. **10.** E: Function F: Not G: Not H: Function. **11.** Function — each moment in time has exactly one height. **12.** Not a function — one score can belong to many students. **13.** C. **14.** (a) F (b) T (c) T. **15.** Yes — every x-value $(2,4,6,8)$ is different, so each input still has exactly one output, even though that output is always 9.

Lesson 9 Assignment *Show your work.*

Part A — Notation and Vocabulary

1. Fill in the blanks: $f(x)$ means the _____ of function _____ when the input is _____.

2. Circle the correct choice in each case.
(a) $f(x)$ means *"f times x" / "the output of f at input x."*
(b) In $f(x)$, the letter x represents the *input / output* of the function.
(c) In $f(x)$, the whole symbol $f(x)$ represents the *input / output* of the function.

3. Rewrite each equation using function notation, with the function named f.
(a) $y = 3x - 2$
(b) $y = x^2 + 1$

Part B — Evaluating for a Numerical Input

4. If $f(x) = x + 8$, determine:
(a) $f(3)$
(b) $f(-6)$
(c) $f(0)$

5. If $f(x) = 4x - 3$, determine:
(a) $f(2)$
(b) $f(-5)$
(c) $f(0)$

6. Function g is defined by $g(x) = 7 - 2x$. Evaluate:
(a) $g(1)$
(b) $g(-3)$
(c) $g(4)$

7. Function h is defined by $h(x) = x^2 - 5$. Evaluate:
(a) $h(3)$
(b) $h(-4)$
(c) $h(0)$

8. A function f is defined by $f(x) = x^2 + 2x$. Find:
(a) $f(2)$
(b) $f(-3)$

9. Consider the functions $f(x) = 4x + 1$ and $g(x) = x^2 - 3$. Evaluate:
(a) $f(5)$
(b) $g(5)$
(c) $f(2) + g(2)$

Part C — Variable and Expression Inputs

10. If $f(x) = 3x + 5$, determine:
(a) $f(a)$
(b) $f(k)$

11. If $f(x) = 5x - 2$, determine an expression for $f(x + 1)$.

12. If $f(x) = x^2 + 6$, determine:
(a) $f(a)$ (b) $f(3a)$

13. If $f(x) = x^2 + 1$, determine a simplified expression for $f(x + 2)$.

Part D — Applications

14. A parking garage charges a flat \$2 entry fee plus \$4 for every hour parked. The total cost, in dollars, of parking for h hours is given by the function $P(h) = 4h + 2$.
(a) Evaluate $P(3)$. Explain, in words, what this value represents.
(b) The statement $P(6) = 26$ is true for this function. Explain, in words, what this statement means.

15. A gym membership costs a \$20 sign-up fee plus \$15 per month. The total cost, in dollars, after t months of membership is given by the function $M(t) = 15t + 20$.
(a) Translate the following sentence into a function notation statement, and check that it is correct: *"The total cost after 5 months is \$95."*
(b) Evaluate $M(10)$, and explain what this value represents.

16. *(Multiple Choice)* A function $C(n)$ gives the cost, in dollars, of buying n candies. Which statement matches $C(6) = 15$?
A. 6 candies cost \$15. B. 15 candies cost \$6. C. Each candy costs \$6. D. \$15 buys 1 candy, with \$6 left over.

Answer Key. 1. output; f; x. **2.** (a) the output of f at input x (b) input (c) output. **3.** (a) $f(x) = 3x - 2$ (b) $f(x) = x^2 + 1$. **4.** (a) 11 (b) 2 (c) 8. **5.** (a) 5 (b) -23 (c) -3. **6.** (a) 5 (b) 13 (c) -1. **7.** (a) 4 (b) 11 (c) -5. **8.** (a) 8 (b) 3. **9.** (a) 21 (b) 22 (c) 10. **10.** (a) $3a + 5$ (b) $3k + 5$. **11.** $5x + 3$. **12.** (a) $a^2 + 6$ (b) $9a^2 + 6$. **13.** $x^2 + 4x + 5$. **14.** (a) 14; parking for 3 hours costs \$14 (b) parking for 6 hours costs \$26. **15.** (a) $M(5) = 95$; since $15(5) + 20 = 95$, this checks out (b) 170; the total cost after 10 months of membership is \$170. **16.** A.

Lesson 10 Assignment *Show your work.*

Part A — Converting Between Notations

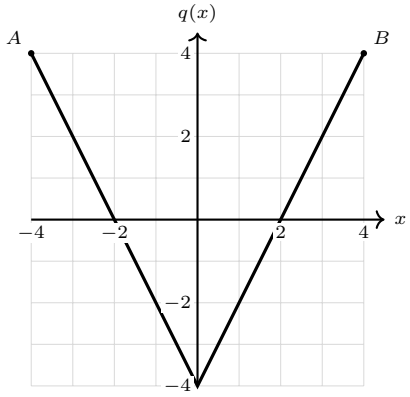
1. Write each function notation equation as an equation in x and y.
(a) $f(x) = 6x - 7$ (b) $g(x) = x + 12$ (c) $Q(t) = 5t - 1$

2. Write each equation in function notation.
(a) $y = 2x + 9$ (b) $y = 15 - 4x$ (c) $y = w - 3$ (name the function g, with input w)

3. Express each equation in function notation.
(a) The graph of the function defined by $y = f(x)$ has equation $y = 1.5x + 4$. (b) The graph of the cost function defined by $y = C(x)$ has equation $y = 3x + 10$ (cost in dollars, for x items).

Part B — Reading Values from Graphs and Tables

4. The graph of a function q is shown at the left.



(a) Complete: **i)** $q(-3) =$ **ii)** $q(1) =$ **iii)** $q(3) =$

(b) Write the ordered pairs associated with i), ii), and iii).

(c) State the value(s) of x if **i)** $q(x) = 4$ **ii)** $q(x) = 0$ **iii)** $q(x) = -4$

(d) State the domain and range of q.

5. The function $g(x) = 2x - 3$ has domain $\{-1, 0, 1, 2, 3\}$. (a) State the range of g. (b) Solve the equation $g(x) = 1$.

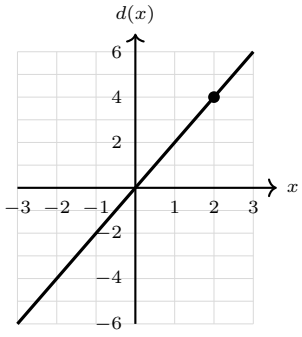
6. The table below shows values of $k(x) = 4x - 5$. (a) Complete the table of values.

x	-1	0	1	2	3
$k(x)$					

(b) State the value of $k(2)$. (c) For what value of x does $k(x) = 11$?

7. Given that $f(x) = 10 - 3x$,
(a) evaluate $f(-4)$ (b) find the value of $f(t) + f(-t)$ (c) determine the x-intercept and y-intercept on the graph of f.

8. The graph of a function d, drawn only for $-3 \leq x \leq 3$, is shown at the left.



(a) A student looks at the marked point and says "$d(4) = 2$." Explain what is wrong with the student's statement, and write the correct statement using function notation.

(b) Solve the equation $d(x) = -6$.

(c) State the domain and range of d.

Part C — Multiple Choice and Numerical Response

9. *(Multiple Choice)* Which statement correctly describes the meaning of the notation $f(5) = 12$?
A. When the output is 5, the input is 12.
B. When the input is 5, the output is 12.
C. The function f is equal to 5 times 12.
D. The point $(12, 5)$ lies on the graph of f.

10. *(Multiple Choice)* Function T is such that $T(6) = -2$. Which of the following statements is correct?
A. The point $(6, -2)$ lies on the graph of $y = T(x)$.
B. The point $(-2, 6)$ lies on the graph of $y = T(x)$.
C. $T(-2) = 6$.
D. Both A and C are correct.

11. *(Numerical Response)* Consider the graph of the function $m(x) = 5x - 20$. The x-intercept of the graph of m is located at $(a, 0)$. The value of a is _____. *(Record your answer in the numerical response box from left to right.)*

Part D — Real-World Function Models

12. Malik earns money mowing lawns. His total earnings are modelled by $E(n) = 15n$, where $E(n)$ is his total earnings, in dollars, for mowing n lawns.

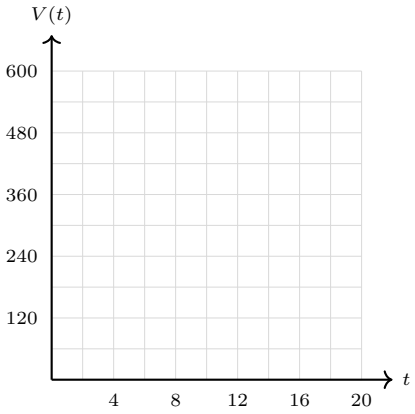
(a) Complete the table of values.

n	1	2	3	4	5
$E(n)$					

(b) Determine the value of $E(7)$, and explain what it represents.

(c) How many lawns did Malik mow if he earned $135?

13. A water tank already contains 40 litres of water when a hose is turned on to fill it further. The volume is modelled by $V(t) = 25t + 40$, where $V(t)$ is the volume, in litres, t minutes after the hose is turned on.



(a) Sketch the graph on the grid at the left, for a maximum of 20 minutes.

(b) Determine the value of $V(10)$.

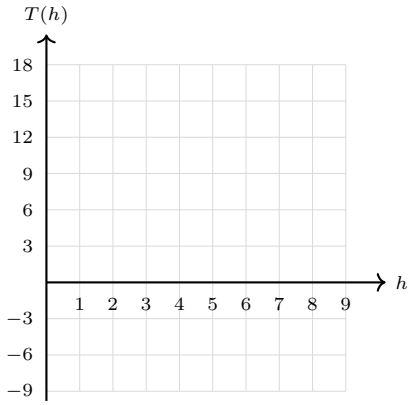
(c) Write in words the meaning of $V(10)$.

(d) Evaluate $V(0)$ and explain its significance.

(e) How many minutes will it take to fill the tank to 540 litres?

(f) Suggest an appropriate domain and range for the function, given that the tank is full at 540 litres.

14. The temperature outside is dropping steadily overnight. It is modelled by $T(h) = 18 - 3h$, where $T(h)$ is the temperature, in degrees Celsius, h hours after 9 pm.



(a) Sketch the graph of the function on the grid at the left.

(b) Determine the value of $T(4)$.

(c) Write in words the meaning of $T(4)$.

(d) Evaluate $T(0)$ and explain its significance.

(e) How many hours after 9 pm will the temperature reach 0 °C?

(f) If recording stops at 6 am (9 hours after 9 pm), suggest an appropriate domain and range for the function.

15. Two ride-share companies charge different rates for a trip. Company A charges $A(k) = 2.5k + 3$ and Company B charges $B(k) = 3k$, where k is the distance travelled, in km, and $A(k)$, $B(k)$ are the costs, in dollars.

(a) Determine the cost of a 10 km trip with each company.

(b) For what distance do both companies charge the same amount? (Solve $A(k) = B(k)$.)

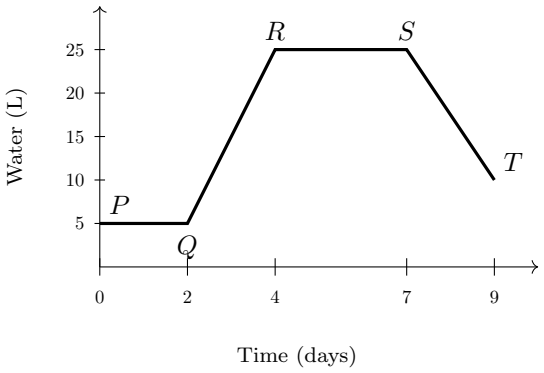
(c) Which company is cheaper for a 20 km trip? Justify your answer using function notation.

Answer Key. 1. (a) $y = 6x - 7$ (b) $y = x + 12$ (c) $y = 5t - 1$. **2.** (a) $f(x) = 2x + 9$ (b) $f(x) = 15 - 4x$ (c) $g(w) = w - 3$. **3.** (a) $f(x) = 1.5x + 4$ (b) $C(x) = 3x + 10$. **4.** (a) $q(-3) = 2$, $q(1) = -2$, $q(3) = 2$ (b) $(-3, 2)$, $(1, -2)$, $(3, 2)$ (c) i) $x = -4, 4$ ii) $x = -2, 2$ iii) $x = 0$ (d) $D : -4 \le x \le 4$; $R : -4 \le q(x) \le 4$. **5.** (a) $R = \{-5, -3, -1, 1, 3\}$ (b) $x = 2$. **6.** (a) $-9, -5, -1, 3, 7$ (b) $k(2) = 3$ (c) $x = 4$. **7.** (a) 22 (b) 20 (c) x-int $= \frac{10}{3}$, y-int $= 10$. **8.** (a) The student swapped input and output; the input is 2 and the output is 4, so $d(2) = 4$ (b) $x = -3$ (c) $D : -3 \le x \le 3$; $R : -6 \le d(x) \le 6$. **9.** B. **10.** A. **11.** 4. **12.** (a) $15, 30, 45, 60, 75$ (b) $E(7) = 105$; Malik earns \$105 for mowing 7 lawns (c) 9 lawns. **13.** (a) line $(0, 40)$ to $(20, 540)$ (b) 290 L (c) at $t = 10$ min the tank holds 290 L (d) $V(0) = 40$; the tank held 40 L before the hose (e) 20 min (f) D: $0 \le t \le 20$, R: $40 \le V(t) \le 540$. **14.** (a) line $(0, 18)$ to $(9, -9)$ (b) 6 °C (c) 4 h after 9 pm it is 6 °C (d) $T(0) = 18$; at 9 pm it was 18 °C (e) 6 h (f) D: $0 \le h \le 9$, R: $-9 \le T(h) \le 18$. **15.** (a) $A(10) = \$28$, $B(10) = \$30$ (b) $k = 6$ km (c) $A(20) = \$53$, $B(20) = \$60$, so Company A is cheaper.

Lesson 11 Assignment *Show your work.*

Part A — Reading and Interpreting Graphs

1. The graph shows the amount of water in a rain barrel, in litres, over a 9-day period, made of four line segments meeting at points P, Q, R, S, T.

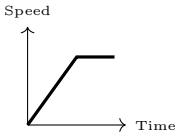


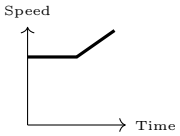
(a) Explain what is happening during each interval: (i) P to Q (ii) Q to R (iii) S to T.

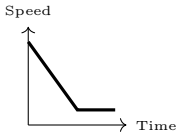
(b) Determine the rate of change, in litres per day, from Q to R.

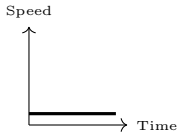
(c) For how many days, in total, was the amount of water constant?

2. *(Multiple Choice)* Which graph below shows a car that speeds up steadily from a complete stop, and then travels at a constant speed?

A.


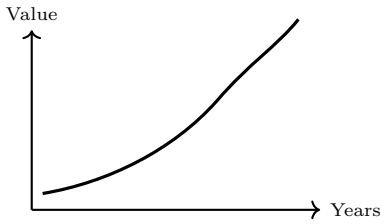
B.


C.


D.


Ans. ___

3. The graph shows the value of a rare coin collection over 6 years (no scale on the axes).

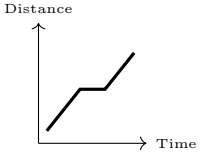


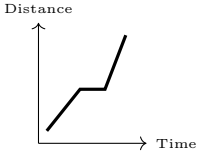
(a) Over the 6 years shown, is the value of the collection increasing, decreasing, or constant?

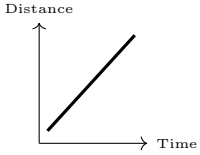
(b) Are the yearly increases in value getting *bigger* or *smaller* as time passes? Explain how you can tell this from the steepness of the graph.

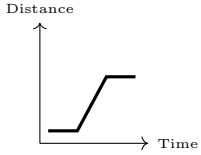
Part B — Matching a Story to a Graph

4. **Story:** Ravi bikes to the park at a steady pace. Partway there, he stops completely to fix his bike chain. Once it's fixed, he bikes the rest of the way to the park at the *same* speed as before. Each graph shows Ravi's distance from home versus time. Which one matches the story?

A.


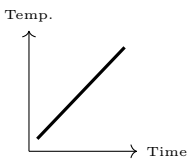
B.


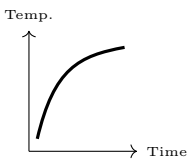
C.


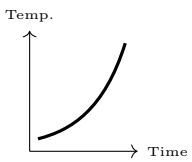
D.


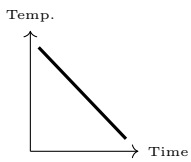
Ans. ___

5. **Story:** The temperature inside a car parked in the sun on a hot day rises quickly at first, then rises more and more slowly as it gets very hot, levelling off. Each graph shows temperature versus time. Which one matches the story?

A.


B.


C.


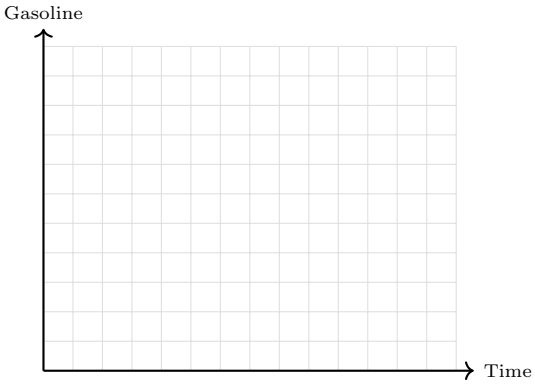
D.


Ans. ___

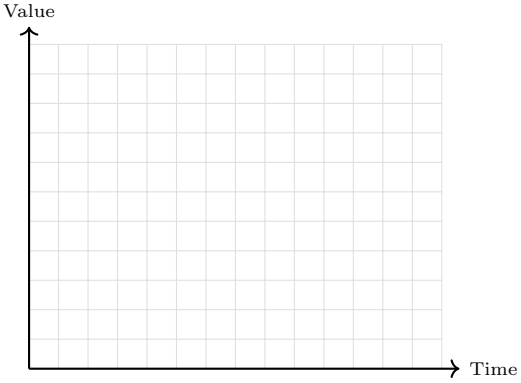
Part C — Sketching a Graph From a Story

Sketch a graph with no scale for each situation. Label your axes.

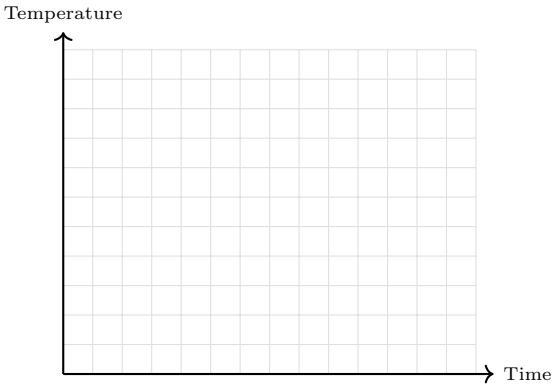
6. The amount of gasoline in a lawnmower's tank, as it is used at a steady rate while mowing, until the tank is completely empty and the mower stops.



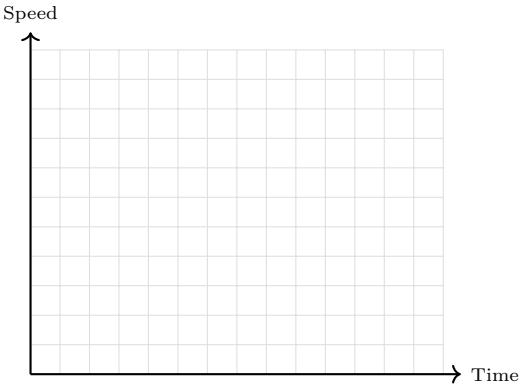
7. The resale value of a new laptop over its first 4 years: it loses value quickly in the first year, then loses value more and more slowly each year after that, levelling off toward a low resale price.



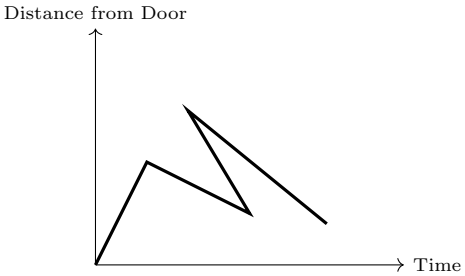
8. The temperature inside a greenhouse from sunrise to sunset: it is coolest at sunrise, warms steadily to a peak in the mid-afternoon, then cools again toward sunset.



9. Your speed while riding an elevator: you start at rest on the ground floor, speed up, travel at a constant speed for several floors, then slow down and come to a stop at the top floor.



Part D — Mixed Practice



10. A student sketched the graph at left to represent his distance from the classroom door during part of a school day. Explain why this graph cannot correctly represent his distance from the door over time.

11. The graph of a hiking trail's elevation versus horizontal distance is made of three line segments meeting at $A(0,\ 200)$, $B(3,\ 500)$, $C(5,\ 500)$, $D(8,\ 50)$ (distance in km, elevation in m).

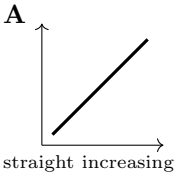
(a) Complete the table.

	A to B	B to C	C to D
Change in distance (km)	3		
Change in elevation (m)	300		
Rate (m/km)	100		

(b) Which segment is the steepest, and what does it represent on the trail?

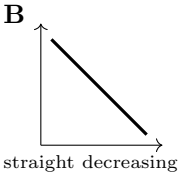
Use the following bank of graph shapes to answer Questions 12–14. Each shape may be used once, more than once, or not at all.

A



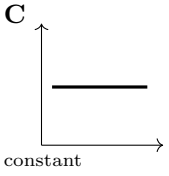
straight increasing

B



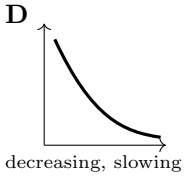
straight decreasing

C



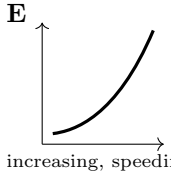
constant

D



decreasing, slowing

E



increasing, speeding

12. The distance remaining on a road trip, if you drive at a constant speed the whole way.
Answer: _____

13. The speed of a hockey puck sliding across rough ice (slows quickly at first, then more and more gently as it comes to a stop).
Answer: _____

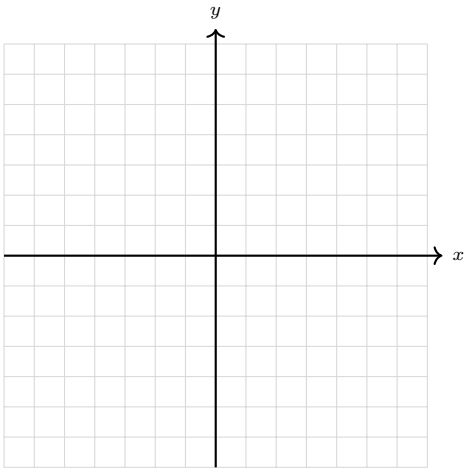
14. The number of views on a video in its first week online, which grows faster and faster each day as more people share it.
Answer: _____

Answer Key. 1. (a)(i) constant 5 L — no rain, none used (ii) rises steadily $5 \to 25$ L (it rains) (iii) falls $25 \to 10$ L (water used). (b) 10 L/day. (c) 5 days (P–Q is 2, R–S is 3). **2.** A. **3.** (a) increasing (b) bigger — the curve gets steeper, so the rate of change is growing. **4.** A. **5.** B. **6.** One straight decreasing segment from full down to 0 (empty). **7.** Decreasing quickly at first, then more and more slowly, levelling off. **8.** Rises steadily to a mid-afternoon peak, then falls toward sunset. **9.** Increases, then flat (constant speed), then decreases back to 0. **10.** The graph crosses itself/turns back in time, so one time would give two distances at once — it fails the vertical line test. **11.** (a) B–C: 2 km, 0 m, 0 m/km; C–D: 3 km, -450 m, -150 m/km. (b) C–D (largest magnitude) — the trail descends sharply back down. **12.** B. **13.** D. **14.** E.

Lesson 12 Assignment *Show your work.*

Part A — The Cartesian Plane

1. State the quadrant containing each point.
(a) $(-2, 9)$ (b) $(7, -3)$ (c) $(-8, -1)$ (d) $(0, -5)$



2. State which axis contains the point $(6, 0)$, and which axis contains the point $(0, -9)$.

3. Plot the points $A(4, 2)$, $B(-3, 5)$, $C(-6, -4)$, $D(2, -7)$ on the grid, and state which quadrant each point lies in.

Part B — Patterns and Representing Relations

4. A stack of firewood grows as logs are added. Level 1 has 3 logs, and each additional level has 4 more logs than the level below it. Complete the table for Levels 1 to 4, and state the number of logs in Level 4.

Level, n	1	2	3	4
Logs, L	3			

5. For the firewood stack in Question 4, identify the independent variable and the dependent variable.

6. The relation is given by the ordered pairs $(1, 4)$, $(2, 8)$, $(3, 12)$, $(4, 16)$. Represent this relation in a table of values.

7. A mapping diagram shows the inputs $\{5, 10, 15\}$ mapped to the outputs $\{1, 2, 3\}$, with $5 \to 1$, $10 \to 2$ and $15 \to 3$. Write this relation as a set of ordered pairs.

Part C — Intercepts, Domain and Range

8. A relation is given by the table below. State the y-intercept and the x-intercept.

x	0	1	2	3	4
y	3	2	1	0	-1

9. The graph of a relation is a line segment with endpoints $(-4, -6)$ and $(2, 5)$. State the domain and range.

10. State the domain and range, using set notation, of the relation $(2, 9)$, $(4, 9)$, $(6, 1)$, $(8, 1)$.

11. State the range of the relation $(-5, 3)$, $(0, 3)$, $(5, 3)$, $(10, 3)$.

Part D — Technology, Functions and Function Notation

12. Priya graphs $y = 6x - 18$ on a graphing calculator, using the standard window $x : [-10, 10]$, $y : [-10, 10]$. Algebraically determine the x-intercept and the y-intercept, and state whether the standard window shows both of them clearly.

13. A line is graphed on a calculator screen and runs off every edge of the window without turning around anywhere. State the domain and range of this relation.

14. Determine whether the relation $(3, 5),\ (4, 5),\ (3, 9),\ (7, 2)$ is a function. Explain.

15. A graph shows the sideways parabola $x = y^2$: it opens to the right from its vertex at the origin, passing through $(1, 1)$, $(1, -1)$, $(4, 2)$ and $(4, -2)$. Is this relation a function? Use the vertical line test to explain your answer.

16. Consider the function $h(x) = -3x + 11$. Determine $h(-2)$.

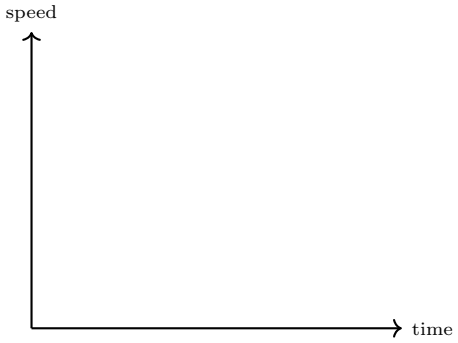
17. For $h(x) = -3x + 11$ from Question 16: if $h(m) = 2$, determine the value of m.

18. The graph of $y = f(x)$ consists of the points $(-4, 6)$, $(-2, 3)$, $(0, 0)$, $(2, -3)$, $(4, -6)$. Determine $f(-2) - f(2)$.

Part E — Models and Real-World Graphs

19. A moving company charges a \$60 flat fee plus \$25 per hour of labour, so the total cost of h hours is $M(h) = 60 + 25h$. Determine $M(3)$, and explain what this value represents.

20. A ball is thrown straight up and lands back on the ground after 4 seconds, reaching its peak height at $t = 2$ seconds. Describe what is happening to the ball's height during the first 2 seconds, and during the last 2 seconds.



21. Sketch a graph showing a car's speed over time as it sits parked (speed 0), accelerates smoothly up to a cruising speed, then holds that cruising speed constant. Describe in words the three sections of your graph.

Part F — Mixed Review

22. A rectangular pool is being filled with water. The volume, in litres, after m minutes is $V(m) = 150m$. (a) Determine $V(12)$. (b) If $V(m) = 2700$, determine how many minutes the pool has been filling.

23. The pool from Question 22 takes 40 minutes to fill completely, after which the hose is turned off. State a reasonable domain for $V(m)$, given this context.

24. A candle is 20 cm tall when lit and burns down at a steady rate of 5 cm per hour. The height, in cm, remaining after x hours is modelled by $f(x) = 20 - 5x$.

(a) Determine $f(3)$, and explain what this value means.

(b) Algebraically determine the x-intercept of this function, and explain what it represents.

(c) State an appropriate domain for this function, given the context.

25. *(Numerical Response)* The function $r(x) = 6x - 4$ has domain $\{1, 2, 3, 4, 5\}$. The sum of all values in the range of r is _____.

26. **Written Response** A parking lot charges according to the function $C(h) = 4 + 3h$, where C is the cost in dollars and h is the number of hours parked, up to a maximum stay of 8 hours.

- Determine the cost of parking for 5 hours.
- If a customer paid \$25, determine how many hours they parked.
- Determine $C(0)$, and explain what this value represents in this situation.
- State the domain and range of this function, given the maximum 8-hour stay.
- The y-intercept of this function's graph equals the value you found in a previous bullet. Explain why this makes sense based on the situation described.

Answer Key. 1. (a) II (b) IV (c) III (d) on the y-axis. **2.** $(6, 0)$: x-axis; $(0, -9)$: y-axis. **3.** A: I, B: II, C: III, D: IV. **4.** 3, 7, 11, 15; Level 4 has 15 logs. **5.** Independent: level number; dependent: number of logs. **6.** x: $1, 2, 3, 4$; y: $4, 8, 12, 16$. **7.** $(5, 1)$, $(10, 2)$, $(15, 3)$. **8.** y-int $(0, 3)$; x-int $(3, 0)$. **9.** D: $-4 \le x \le 2$; R: $-6 \le y \le 5$. **10.** $D = \{2, 4, 6, 8\}$, $R = \{1, 9\}$. **11.** $R = \{3\}$. **12.** x-int $(3, 0)$; y-int $(0, -18)$ — below Ymin $= -10$, so *not* shown clearly. **13.** Domain and range are both all real numbers. **14.** Not a function — $x = 3$ pairs with both 5 and 9. **15.** Not a function — the vertical line $x = 4$ crosses it at $(4, 2)$ and $(4, -2)$. **16.** 17. **17.** $m = 3$. **18.** $3 - (-3) = 6$. **19.** $M(3) = \$135$: the total cost of a 3-hour move is \$135 (flat fee plus 3 hours of labour). **20.** Height rises for the first 2 s (up to the peak), then falls for the last 2 s (back to the ground). **21.** Flat at 0 (parked), rising (accelerating), then flat at a higher value (cruising). **22.** (a) 1800 L (b) $m = 18$ min. **23.** $0 \le m \le 40$. **24.** (a) $f(3) = 5$: 5 cm of candle remains after 3 h (b) $(4, 0)$: the candle burns out after 4 h (c) $0 \le x \le 4$. **25.** 70. **26.** $C(5) = \$19$; $h = 7$ h; $C(0) = \$4$, the base fee charged even for 0 hours; D: $0 \le h \le 8$, R: $\$4 \le C \le \$28$; the y-intercept is the cost at $h = 0$ — the same base fee, before any hourly charge applies.